

NIKTA SHAMS-MOHAMMADI

Ph.D. Candidate, Salerno, Italy

 Email  LinkedIn  Google Scholar

SUMMARY

My research focuses on the development of mathematical and computational models for the analysis and simulation of complex physical systems under real world conditions. In particular, my recent study integrates physics based modeling with machine learning techniques to address inverse problems and parameter estimation in nonlinear systems. In addition, I explore image/signal processing and computer vision methods for extracting meaningful features, enabling data driven analysis, pattern recognition, and enhanced system interpretation in complex physical problems.

EDUCATION

Università degli Studi di Salerno

Salerno, Italy

Ph.D. in Photovoltaics – Design and Integration

Nov 2022 – Nov 2025

- National PhD Program in Photovoltaics (Awarded Unisa scholarship)
 - Thesis: Physics-Informed Neural Network Modeling for Intelligent Health Monitoring and Parameter Identification of Nonlinear Photovoltaic Systems
- Supervisor: [Prof. Giovanni Spagnuolo](#)

Shahrood University of Technology

Shahrood, Iran

M.Sc. in Electrical Engineering – Telecommunication Systems

Oct 2020 – Jul 2022

GPA: 4.00/4.00 (Ranked 1st)

- Thesis: Improving License Plate Image Resolution Using a Deep Learning Framework Based on U-Net
- Supervisor: [Prof. Alireza Ahmadyfard](#)

Golestan University

Gorgan, Iran

B.Sc. in Electrical Engineering

Sep 2016 – Sep 2020

GPA: 3.02/4.00 (15.58/20)

- Thesis: Computational Framework for EEG-Based Epilepsy Diagnosis
- Supervisor: [Prof. Mohammadhasan Miri](#)

RESEARCH INTERESTS

- Physics-Informed Neural Networks (PINNs) for modeling and parameter estimation
- Dynamic Modeling, System Identification, and Control of nonlinear and complex systems
- Image and Signal Processing for feature extraction, and pattern recognition
- Machine Learning and Deep Learning techniques for data-driven modeling and predictive analysis

EXPERIENCE

• Visiting Researcher Experience

Universidad de Málaga

Málaga, Spain

Visiting PhD Student

Feb 2025 – Sep 2025

Designed and developed two novel physics-informed neural network (PINN) framework (Two level layered and Multiple PINN) for dynamic photovoltaic system modeling, enabling robust parameter estimation under noisy and real-world operating conditions.

Supervisors: [Prof. José del Campo-Ávila](#), [Prof. Llanos Mora López](#), [Prof. Esteban Palomo](#)

• Teaching Experience

Shahrood University of Technology

Shahrood, Iran

Teaching Assistant – Signals and Systems

2022

Assisted in teaching signal analysis and systems to over 40 undergraduate students, connecting theoretical concepts with practical engineering examples.

Mentored students through problem-solving sessions and guided assignments, supporting measurable improvement in conceptual understanding and course outcomes.

Supervisor: [Prof. Hossein Marvi](#)

• Internship Experience

Aliabad Gas Power Plant

Gorgan, Iran

Undergraduate Internship – Gas Power Plant

2019

Gained hands-on exposure to operation and maintenance activities of gas power plant systems, including inspection procedures and support in repair processes.

Developed practical understanding of electrical and control components within large-scale power generation infrastructure under industrial safety regulations.

Supervisor: [Prof. Alireza Safa](#)

• Collaboration Experience

Universidad de Málaga

Málaga, Spain

Informatics Department

2025

Designed and implemented convolutional neural network (CNN)-based models for fault detection and diagnostics of photovoltaic panels using image data.

Focused on feature learning and classification of defect patterns to improve fault recognition accuracy under real operating conditions.

Supervisors: [Prof. José del Campo-Ávila](#), [Prof. Esteban Palomo](#)

University of Calabria

Calabria, Italy

Civil Engineering Department

2022

Applied U-Net-based convolutional neural networks for image enhancement of traffic image datasets, targeting improved visual quality for smart parking applications.

Focused on learning-based enhancement under varying illumination and noise conditions to support downstream detection tasks.

Supervisors: [Prof. Giuseppe Guido](#), [Prof. Vittorio Astarita](#)

Babol Noshirvani University of Technology

Babol, Iran

Mechanical Engineering Department

2021

Modeled and analyzed nonlinear dynamical systems using analytical and numerical techniques.

Applied Runge-Kutta numerical analysis and semi-analytical methods analysis such as the Akbari-Ganji Method (AGM) and Homotopy Perturbation Method (HPM) to analysis of nonlinear dynamics system.

Supervisor: [Prof. Davood Domiri Ganji](#)

Golestan University

Gorgan, Iran

Mechanical Engineering Department

2019

Modeled and analyzed a piezoelectric energy harvesting cantilever, incorporating geometric configuration, material properties, and electromechanical coupling effects.

Supervisor: [Prof. Mohammad Rahimzadeh](#)

PUBLICATIONS

• Journal papers

Self-Adaptive Loss Weighting in Physics-Informed Neural Networks for Photovoltaics Dynamic Model Identification 2025

Shamsmohammadi, N., Spagnuolo, G., J. del Campo-Ávila, E. J. Palomo, and L. Mora-López

under review

Enhanced Parameter Estimation in Photovoltaic Models via Multiple Physics-Informed Neural Networks 2025

Shamsmohammadi, N., Spagnuolo, G., J. del Campo-Ávila, E. J. Palomo, and L. Mora-López

under review

Estimation of Parameters in Photovoltaic Models Using Two-Level Layered Physics-Informed Neural Network 2025

Shamsmohammadi, N., Spagnuolo, G., J. del Campo-Ávila, E. J. Palomo, and L. Mora-López

Journal of Photovoltaics

Investigating and Classifying the Level of Thermal Runaway-Induced Failure in Energy Storage Systems of Photovoltaics 2024

Shamsmohammadi, N.

Journal of Computational and Cognitive Engineering

Enhancing Operational Efficacy of Smart Parking Facilities through Intelligent License Plate Recognition 2024

Shamsmohammadi, N., Shaffiee Haghshenas, S., Guido, G., and Astarita, V.

Journal of Soft Computing in Civil Engineering

Improving the Efficiency of a Cantilever Energy Scavenger 2023

Rahimzadeh, M., Samadi, H., Shamsmohammadi, N.

Journal of Renewable Energy and Environment

Nano/micro-beam Deflections: Investigation of Subjected Forces and Applications 2023

Shamsmohammadi, N., Samadi, H., Rahimzadeh, M., Asadi, Z., Ganji, D.D.

Physics Open

Analytical Investigation and Comparison of Oscillating Systems with Nonlinear Behavior Using AGM and HPM 2022

Samadi, H., Mohammadi, N.S., Shamoushaki, M., Asadi, Z., Ganji, D.D.

Alexandria Engineering Journal

Analysis of Energy Harvesting Enhancement in Piezoelectric Unimorph Cantilevers 2021

Rahimzadeh, M., Samadi, H., Shamsmohammadi, N.

Sensors

• Conference papers

Photovoltaic Module Single-Diode-Model Identification by a Physics-Informed Neural Network 2025

Shamsmohammadi, N., Spagnuolo, G.

International Joint Conference on Neural Networks (IJCNN)

(Selected for extension and submission to IEEE Transactions on Neural Networks and Learning Systems)

Parametric Identification of the Dynamic Photovoltaic Model by a Physics-Informed Neural Network 2025

Shamsmohammadi, N., Spagnuolo, G.

International Conference on Clean Electrical Power (ICCEP)

ACTIVITIES

• Memberships

Member of the International Neural Network Society (INNS)

Member of the Institute of Electrical and Electronics Engineers (IEEE)

Member of the Electrical Engineering Scientific Association, University of Golestan, Iran

• Reviewer Activities

Journal of Computational and Cognitive Engineering (3)
International Joint Conference on Neural Networks (IJCNN) (2)
Journal of Soft Computing in Civil Engineering (2)
Ocean Engineering (1)
Journal of Process Mechanical Engineering (1)

• Organizing Committee Member

5th Iranian Conference on Communications Engineering (ICCE 2021)

HONORS AND AWARDS

- Awarded the UNISA Scholarship as the top-ranked international applicant based on academic merit.
- Ranked 1st among all M.Sc. graduates in Electrical Engineering in class.
- The paper "Nano/micro-beam deflections: Investigation of subjected forces and applications" was selected to publish in the *Science Featured Series*.

REFERENCES

Prof. Giovanni Spagnuolo

Full Professor, University of Salerno

Department of Information and Electrical Engineering and Applied Mathematics (DIEM), Fisciano (SA), Italy

[Email](#) | [Website](#) | [Google Scholar](#)

Prof. José del Campo Ávila

Associate Professor, Universidad de Málaga

School of Computer Engineering, Campus de Teatinos, Málaga, Spain

[Email](#) | [Website](#) | [Google Scholar](#)